

Analysis of Clocked Sequential Circuits

Analysis of Clocked Sequential Circuits

- Analysis describes the behaviour of a clocked sequential circuits under certain operating conditions.
- It can be determined from the **inputs, outputs and state of flip flops**.
- The outputs and the next state are both a function of the inputs and the present state.
- The analysis of a sequential circuit obtained by **state table** or a **state diagram**.
- It is also possible to write Boolean expressions that describes the behaviour of the sequential circuits.
- A logic diagram is recognized as a clocked sequential circuit it includes flip flops with clock inputs.
- The FF may be of any type and the logic diagram may or may not include combinational logic gates.

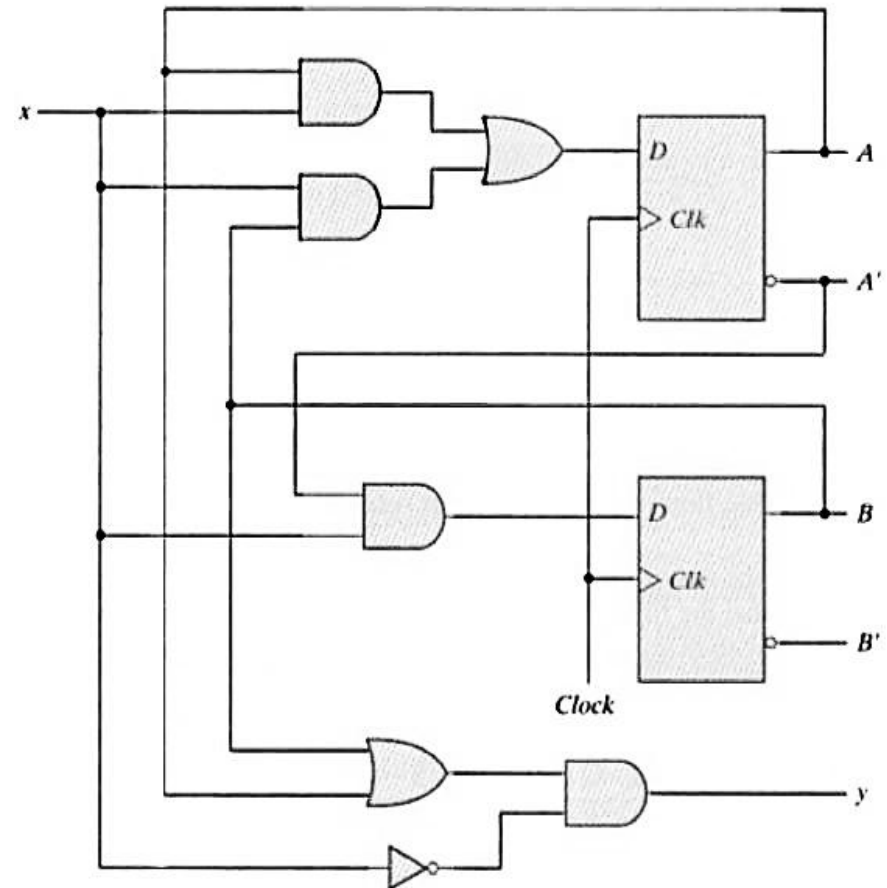
Introduction to State equation, State table and State diagram

- The behaviour of a clocked sequential circuit can be described by means of state equations.
- A state equation (transition equation) specifies the next state as a function of the present state and inputs.

$$A+ = A(t+1) = Ax + Bx$$

$$B+ = B(t+1) = A'x$$

$$y = (A+B)x'$$



- **State Table:** The time sequence of inputs, outputs and flip flops can be enumerated in a state table.
- It consists of four sections such as present state, input, next state, and output.

$$A^+ = A(t+1) = Ax + Bx$$

$$B^+ = B(t+1) = A'x$$

$$y = (A+B)x'$$

Present State			Next State		Output
<i>A</i>	<i>B</i>	<i>x</i>	<i>A</i> ⁺	<i>B</i> ⁺	<i>y</i>
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

- **State Diagram:** The information available in a state table can be represented graphically in the form of a state diagram.
- A state is represented by a circle and the transitions between states are indicated by directed lines connecting the circles.

- **State assignments:**

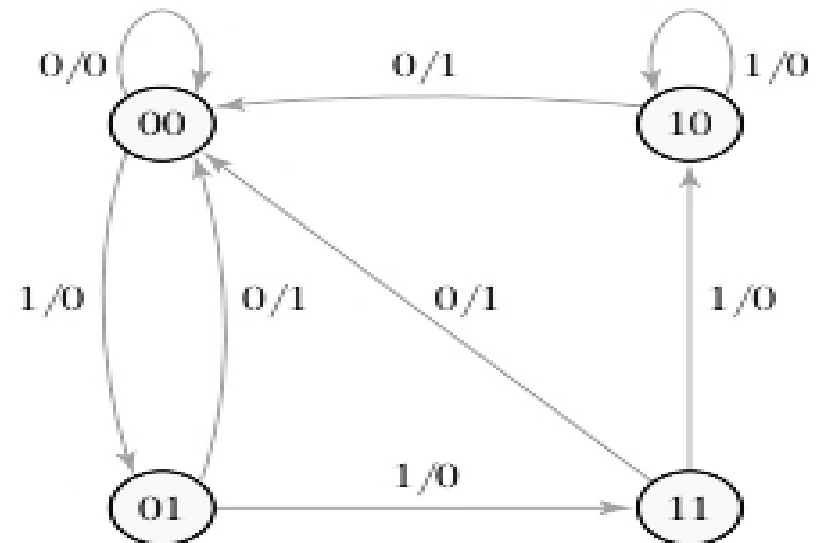
S0 = 0 0

S1 = 0 1

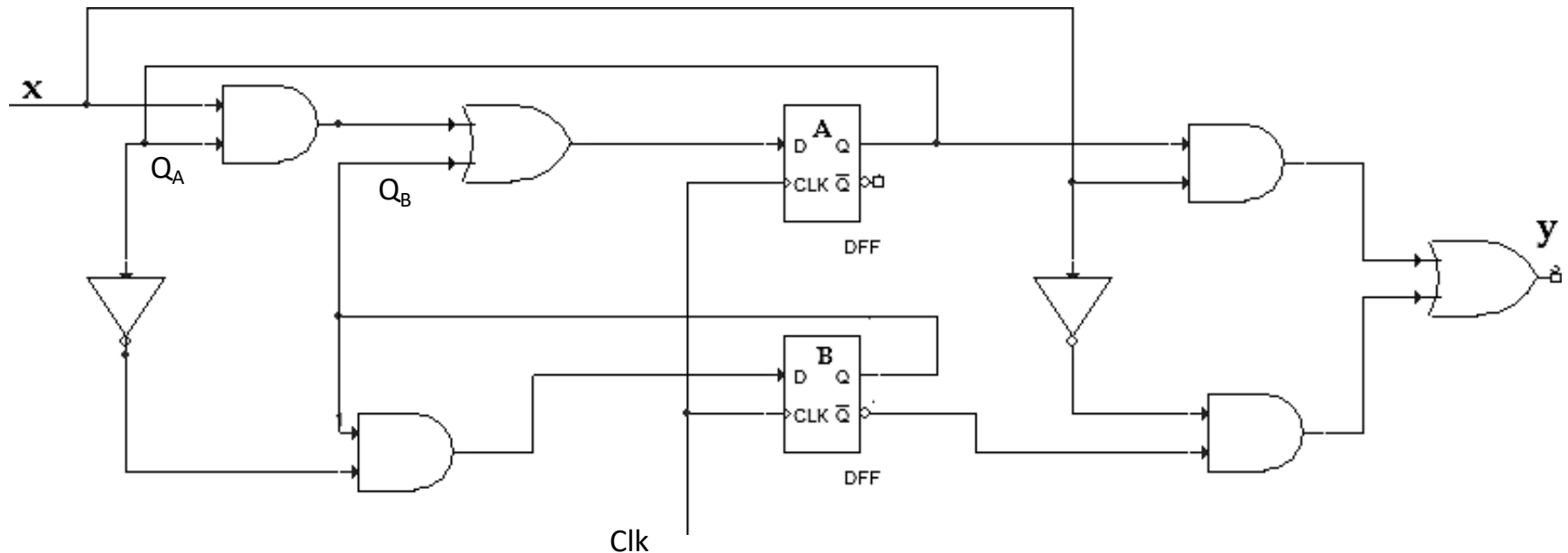
S2 = 1 0

S3 = 1 1

Present State		Input	Next State		Output
A	B		A ⁺	B ⁺	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0



Analysis of Clocked Sequential Circuit with D Flip Flop



Step 1: Find the input and output equations:

$$D_A = X Q_A + Q_B$$

$$D_B = Q'_A Q_B$$

$$Y = X Q_A + X' Q'_B$$

Characteristic equation:

$$Q(t+1) = D$$

Next State:

$$Q_{A+} = D_A = X Q_A + Q_B$$

$$Q_{B+} = D_B = Q'_A Q_B$$

$$Y = X Q_A + X' Q'_B$$

State Table:

Present State		Input X	Next State		Output Y
Q_A	Q_B		Q_{A+}	Q_{B+}	
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	1
1	1	0	1	0	0
1	1	1	1	0	1

State diagram: 2 FF, hence 4 states

Q_A Q_B

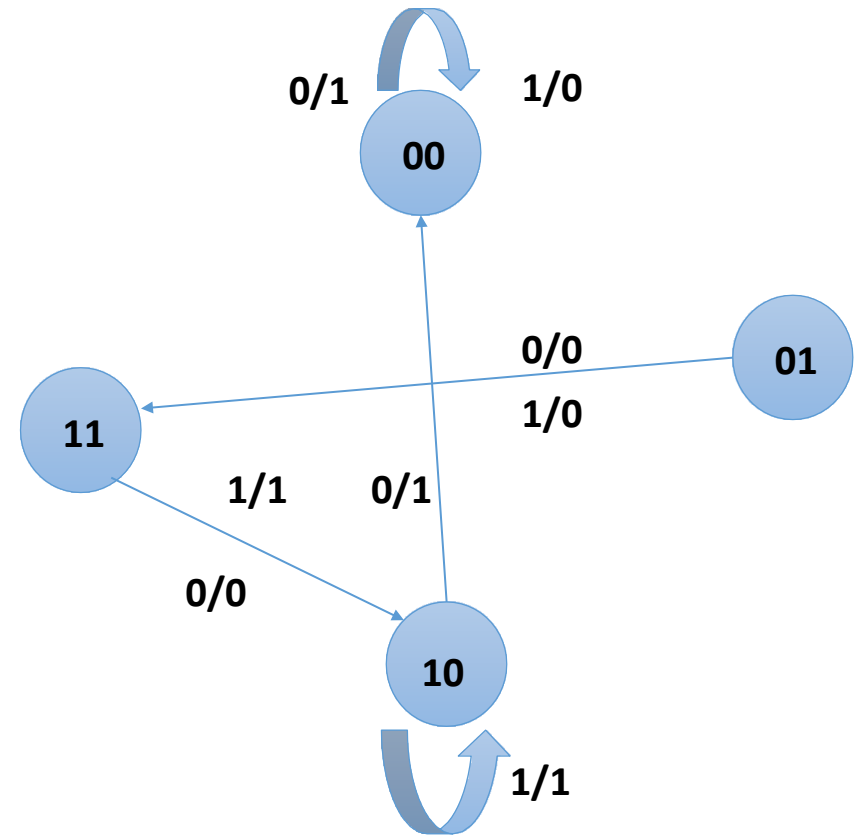
S0 = 0 0

S1 = 0 1

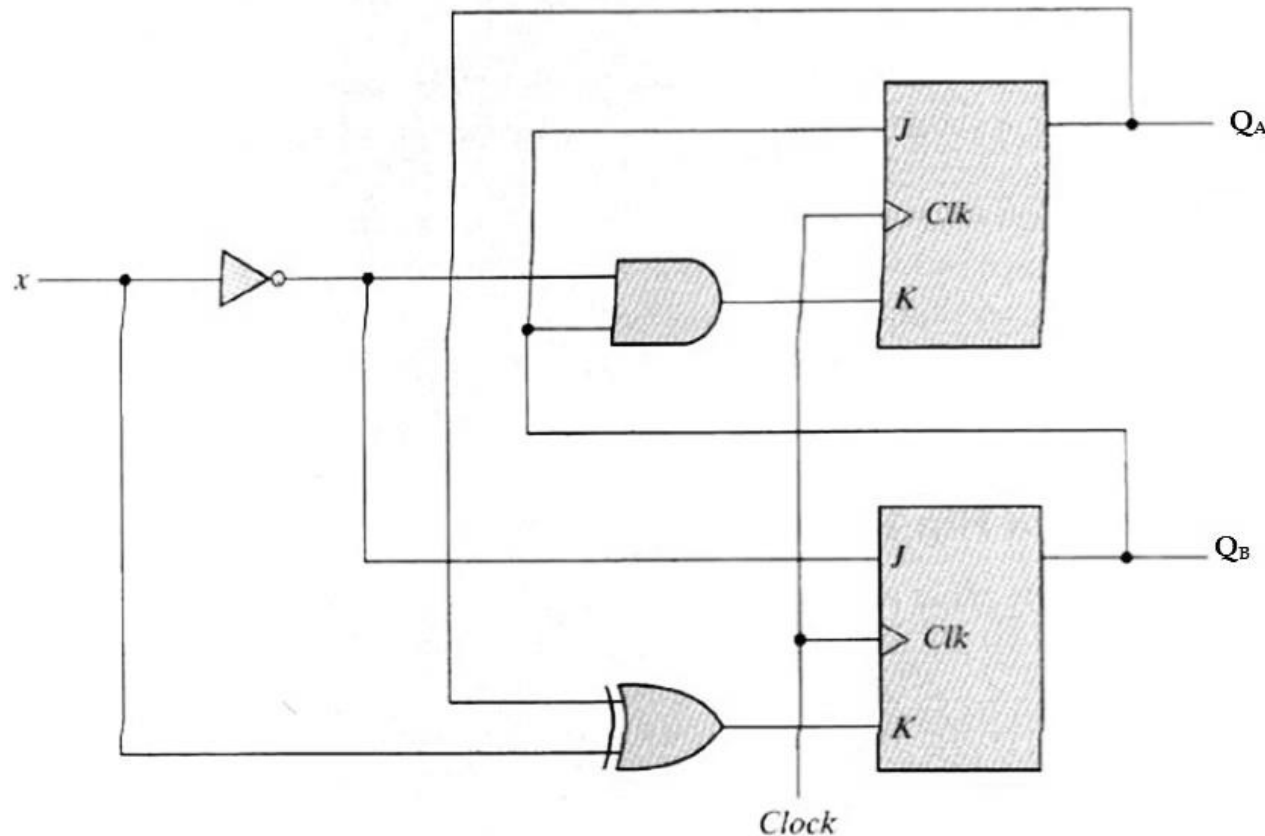
S2 = 1 0

S3 = 1 1

Present State		Input X	Next State		Output Y
Q_A	Q_B		Q_{A+}	Q_{B+}	
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	1
1	1	0	1	0	0
1	1	1	1	0	1



Analysis of Clocked Sequential Circuit with JK Flip Flop



- $J_A = Q_B$
- $K_A = X' Q_B = X' J_A$
- $J_B = X'$
- $K_B = X \oplus Q_A$

Step 1: Find the input and output equation.

- There is no output combinational logic, hence no output equation.
- $J_A = Q_B$
- $K_A = X' Q_B = X' J_A$
- $J_B = X'$
- $K_B = X \oplus Q_A$

- Characteristic table and equation:

J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	Q(t)'

$$Q(t+1) = J.Q(t)' + K'.Q(t)$$

- $J_A = Q_B$
- $K_A = X' Q_B = X' J_A$
- $J_B = X'$
- $K_B = X \oplus Q_A$

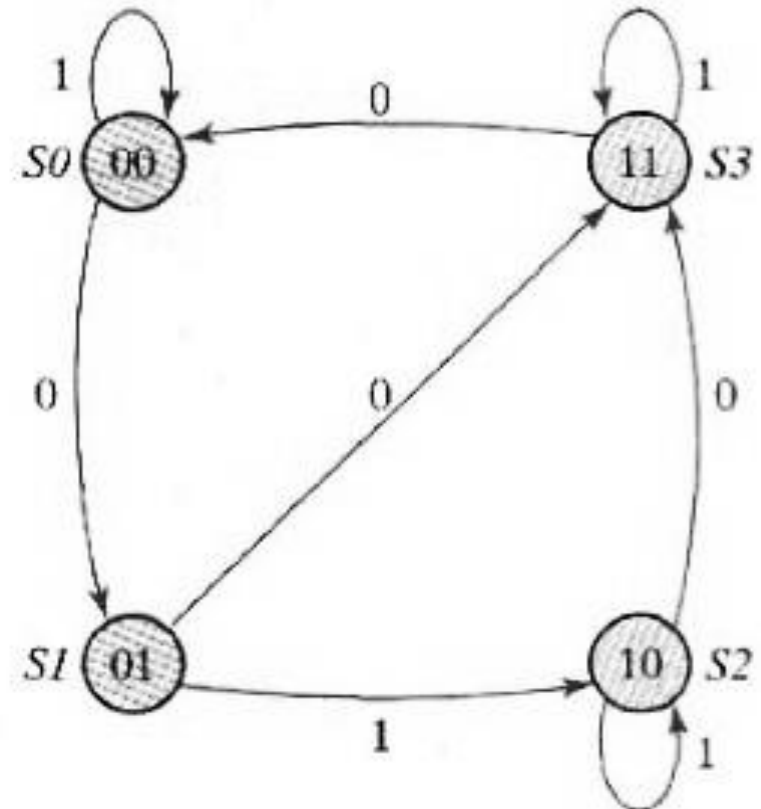
• **Step 2: State Table**

$$Q(t+1) = J.Q(t)' + K'.Q(t)$$

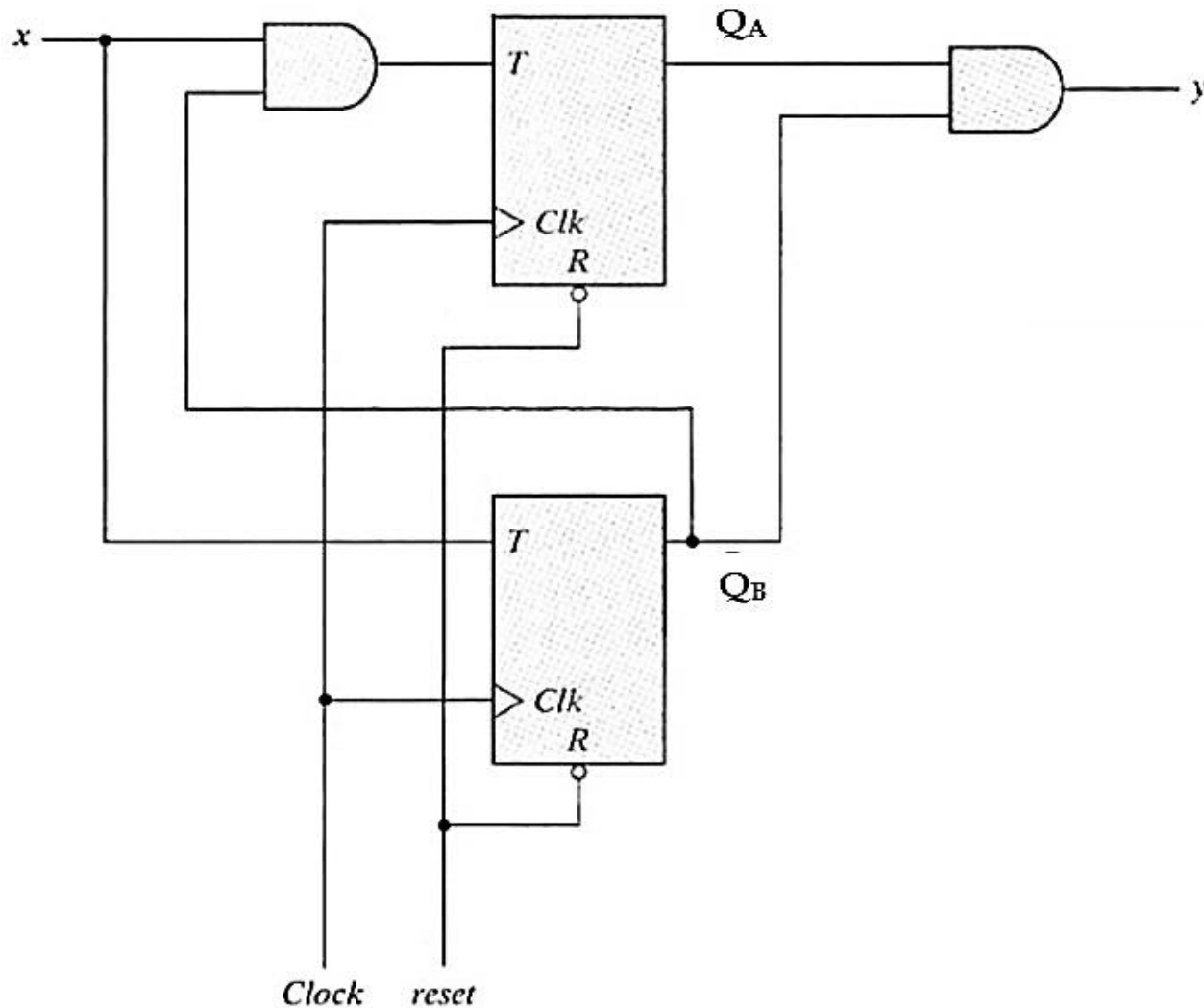
Present State		Input	Flip-Flop Inputs				Next State	
Q_A	Q_B		J_A	K_A	J_B	K_B	Q_A^+	Q_B^+
0	0	0	0	0	1	0	0	1
0	0	1	0	0	0	1	0	0
0	1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	1	0
1	0	0	0	0	1	1	1	1
1	0	1	0	0	0	0	1	0
1	1	0	1	1	1	1	0	0
1	1	1	1	0	0	0	1	1

- Step 3: State Diagram.

Present State		Input	Next State	
Q_A	Q_B		Q_A^+	Q_B^+
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1



Analysis of Clocked Sequential Circuit with T Flip Flop



- **Step 1: Find the input and output equation.**

- $T_A = X Q_B$

- $T_B = X$

- $Y = Q_A Q_B$

- **Characteristic table and equation:**

T Flip-Flop		
T	Q(t + 1)	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

$$Q_{A+} = Q_A \oplus T_A$$

$$Q_{B+} = Q_B \oplus T_B$$

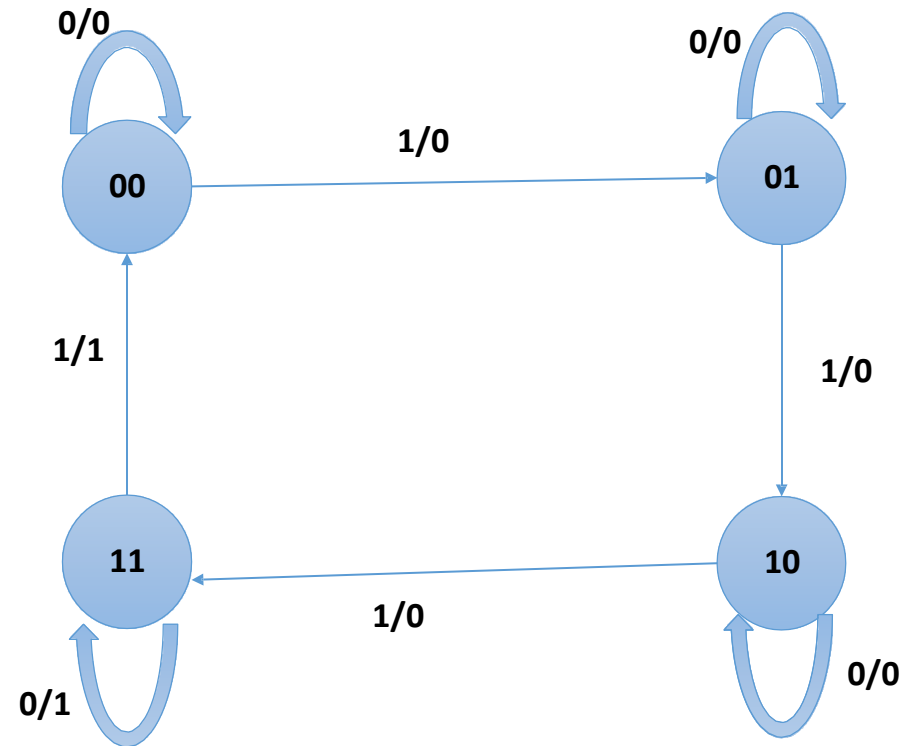
$$Q(t+1) = T.Q(t)' + T'.Q(t)$$

• **Step 2: State Table.**

Present State		Input			Next State		Output
Q_A	Q_B	X	T_A	T_B	Q_A^+	Q_B^+	Y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	0	0	0	1	0
0	1	1	1	1	1	0	0
1	0	0	0	0	1	0	0
1	0	1	0	1	1	1	0
1	1	0	0	0	1	1	1
1	1	1	1	1	0	0	1

• Step 3: State Diagram.

Present State		Input X	T_A T_B		Next State		Output Y
Q_A	Q_B				Q_{A+}	Q_{B+}	
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	0	0	0	1	0
0	1	1	1	1	1	0	0
1	0	0	0	0	1	0	0
1	0	1	0	1	1	1	0
1	1	0	0	0	1	1	1
1	1	1	1	1	0	0	1



Counters

- A register that goes through a prescribed sequence of states upon the application of input pulses (clock pulse) is called a counter.
- Two main types of counters
 - Asynchronous (Ripple) counters
 - Synchronous counters

Asynchronous (Ripple) counter

- The flip-flop output transition serves as a source for triggering other flip-flops
- Hardware is much simpler
- Unreliable and delay dependent due to the asynchronous behaviors

Synchronous counter

- The **common clock pulse** triggers all flip-flops simultaneously
- Hardware may be more complex but more reliable
- Synchronous counters are more widely used in industry.

Difference

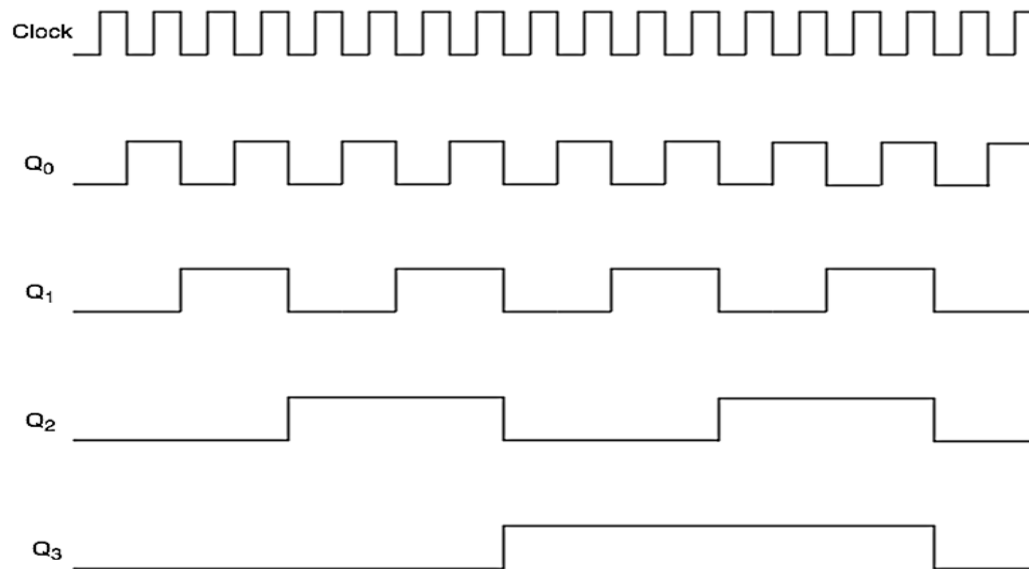
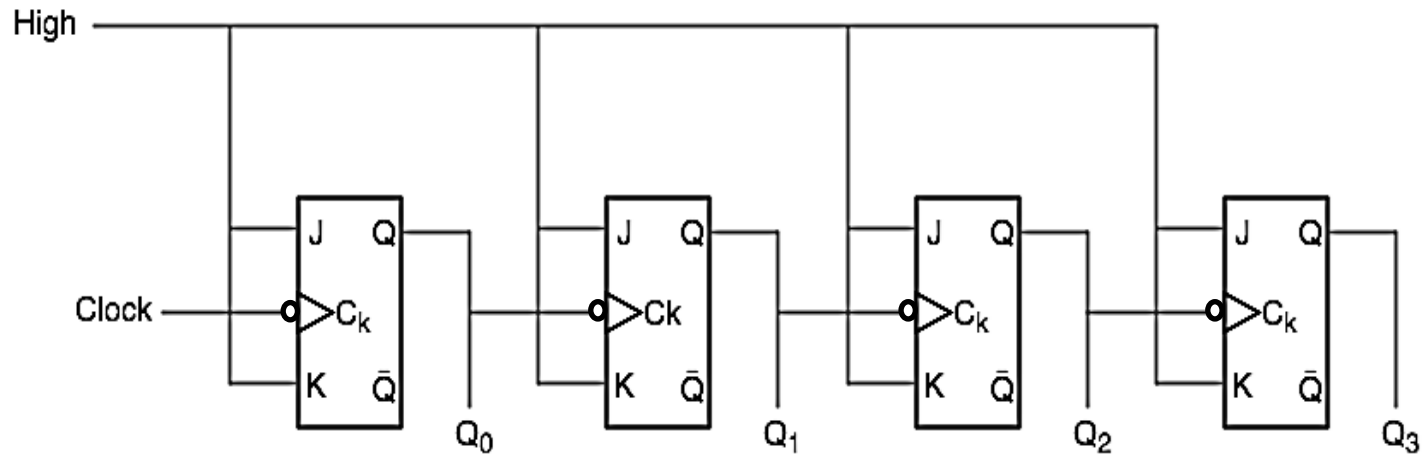
Synchronous Counter

- In synchronous counter we use a universal clock that is common to all flip flops through out the circuit.
- It works faster.
- Called as parallel Counter.
- Synchronous Counter design as well implementation are complex due to increasing number of states.
- Synchronous Counter will operate in any desired count sequence.
- Propagation delay is less.

Asynchronous Counter

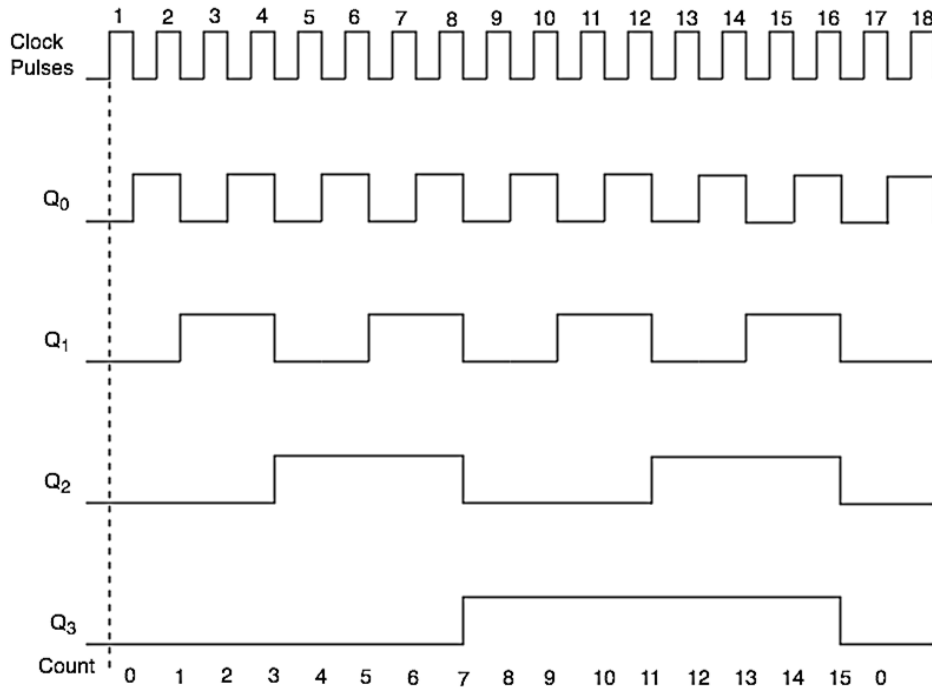
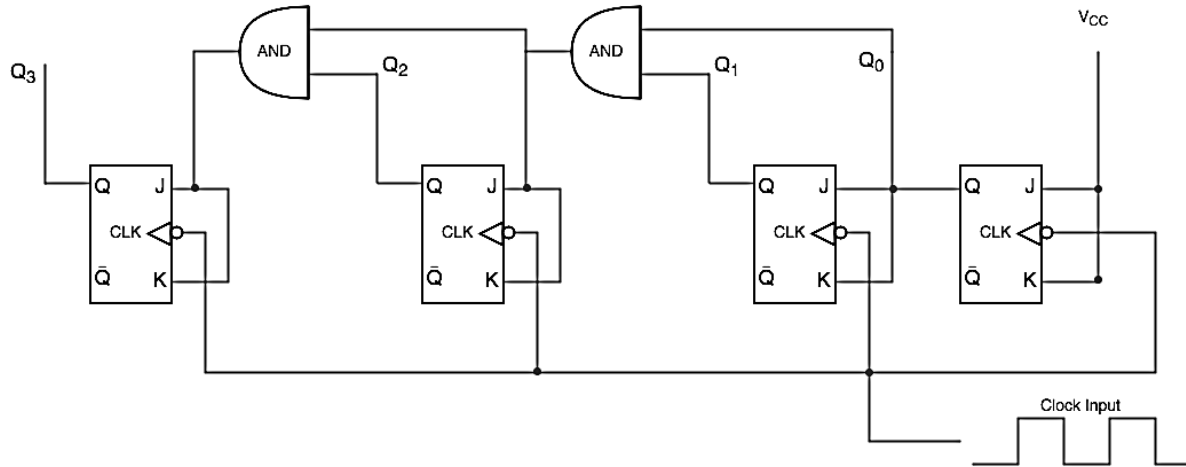
- In asynchronous counter main clock is only applied to the first flip flop and then for rest of flip flops the output of previous flip flop is taken as a clock.
- It works slower compared to synchronous counter.
- Called serial Counter.
- Asynchronous Counter designing as well as implementation is very easy.
- Asynchronous Counter will operate only in fixed count sequence (UP/DOWN).
- Propagation delay is more.

Example of Asynchronous Counter



(b) Timing Diagram

Example of Synchronous Counter



Design of Counters

- Counters can be designed to generate any desired sequence of states.
- A divide-by- N counter (**mod - N counter**) is a counter that goes through a repeated sequence of **N states**.
- **Mod - N counter** Where N represents the no.of states.
- **$N=2^n$** where n represents the no.of flip flops needed
- The sequence may follow the binary count or may be any other arbitrary sequence.
- Counters can also be constructed by means of shift registers.

Steps to Design of Synchronous Counters

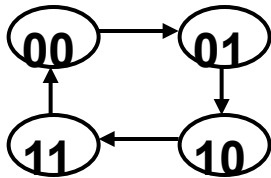
1. Number of Flip flops required
2. State diagram
3. Choice of flip flops & Excitation Table
4. Minimal expression
5. Logic (circuit) diagram

Synchronous (Parallel) 2 bit Up Counter Or Mod - 4 counter

- Synchronous (parallel) counters: the flip-flops are clocked at the same time by a common clock pulse.

1. Number of flipflops = number of bits = 2

2. State Diagram



3. Excitation table

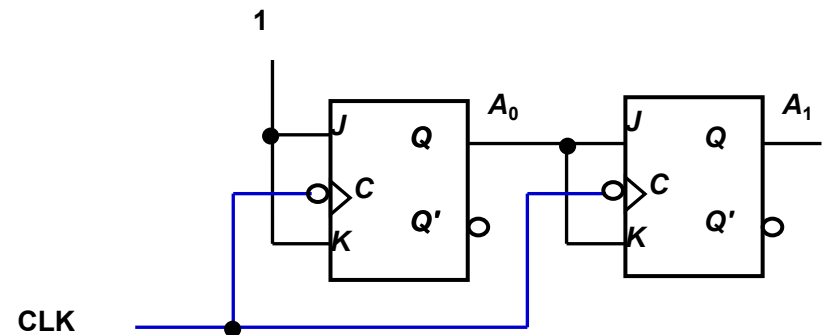
Present state		Next state		Flip-flop inputs	
A_1	A_0	A_1^+	A_0^+	TA_1	TA_0
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	0	1	1

4. Minimal Expression

$$TA_1 = A_0$$

$$TA_0 = 1$$

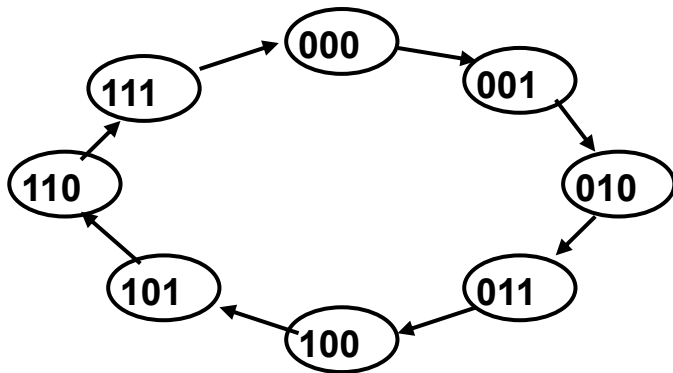
5. Circuit diagram



Synchronous (Parallel) 3 bit Up Counter Or Mod - 8 counter

1. Number of flipflops = number of bits = 3 (using T flip-flops, or JK flip-flops with identical J, K inputs).

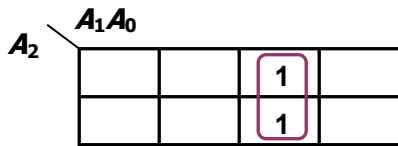
2. State Diagram



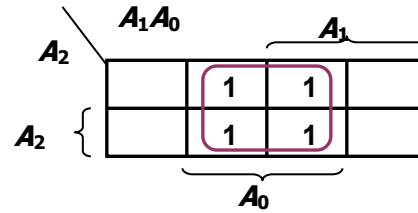
3. Excitation table

Present state			Next state			Flip-flop inputs		
A_2	A_1	A_0	A_2^+	A_1^+	A_0^+	TA_2	TA_1	TA_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

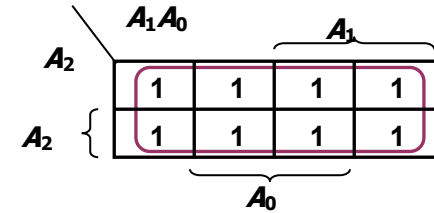
4. Minimal Expression



$$TA_2 = A_1 \cdot A_0$$

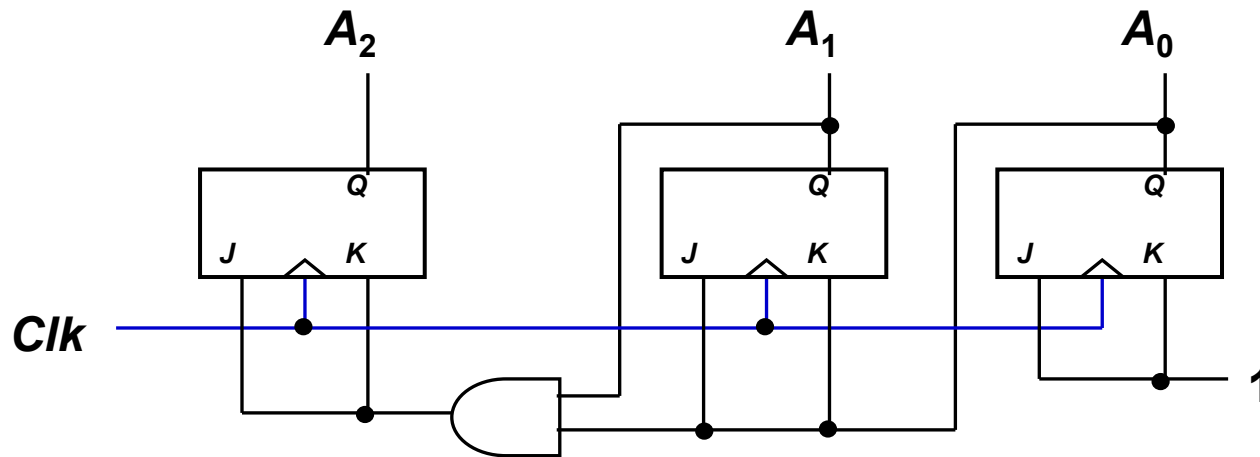


$$TA_1 = A_0$$



$$TA_0 = 1$$

5. Circuit diagram



Synchronous (Parallel) n-bit Counter

- Note that in a binary counter, the n^{th} bit (shown underlined) is always complemented whenever
 - $\underline{0}11\dots11 \rightarrow \underline{1}00\dots00$
 - or $\underline{1}11\dots11 \rightarrow \underline{0}00\dots00$
- Hence, X_n is complemented whenever
- $X_{n-1}X_{n-2} \dots X_1X_0 = 11\dots11.$
- As a result, if T flip-flops are used, then
- $TX_n = X_{n-1} \cdot X_{n-2} \cdot \dots \cdot X_1 \cdot X_0$

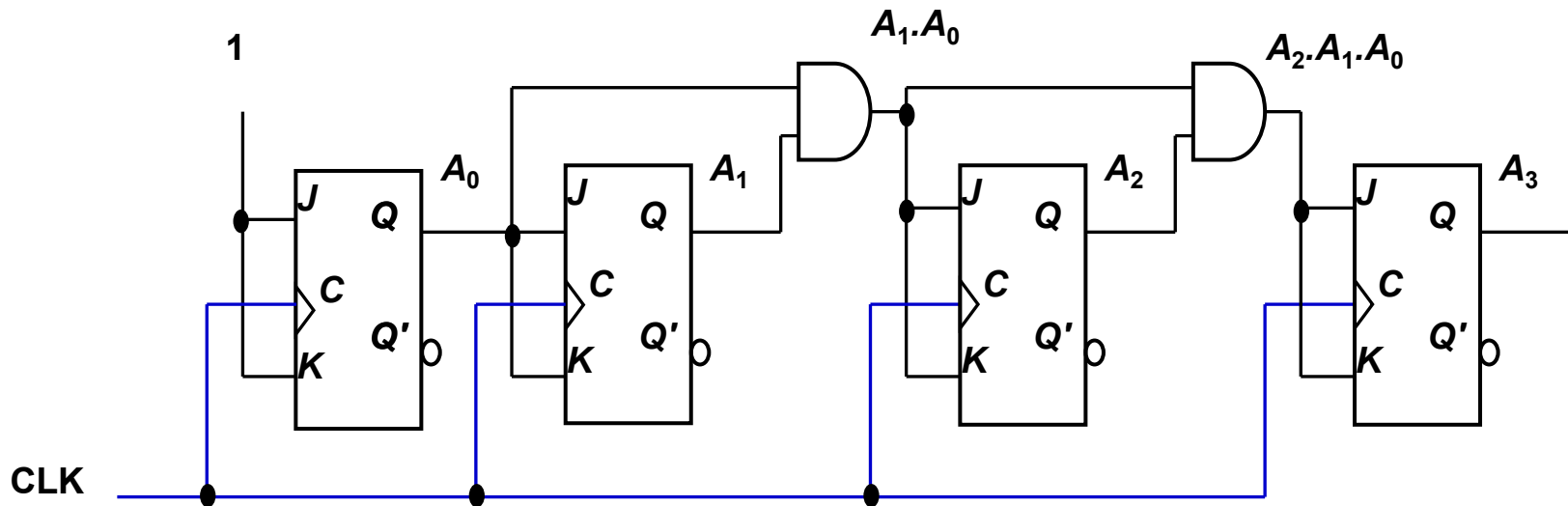
Synchronous (Parallel) 4-bit Up Counter Or Mod - 16 counter

$$TA_3 = A_2 \cdot A_1 \cdot A_0$$

$$TA_2 = A_1 \cdot A_0$$

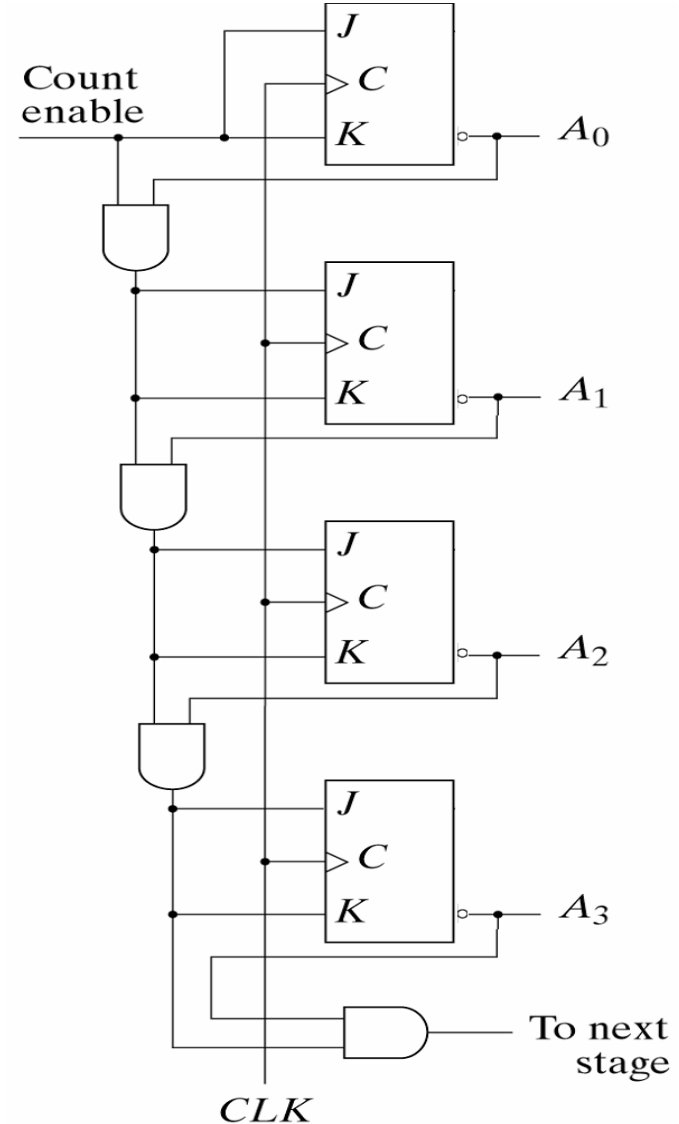
$$TA_1 = A_0$$

$$TA_0 = 1$$



Synchronous (Parallel) 4-bit Down Counter Or Mod - 16 counter

A_3	A_2	A_1	A_0
1	1	1	1
1	1	1	0
1	1	0	1
1	1	0	0
1	0	1	1
1	0	1	0
1	0	0	1
1	0	0	0
0	1	1	1
0	1	1	0
0	1	0	1
0	1	0	0
0	0	1	1
0	0	1	0
0	0	0	1
0	0	0	0



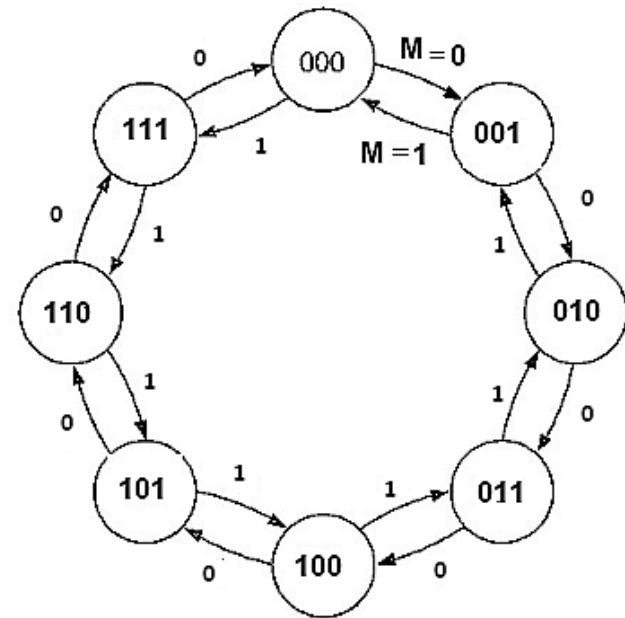
Up/Down Synchronous Counters

- Up/down synchronous counter: a *bidirectional* counter that is capable of counting either up or down.
- An input (control) line *Up/Down* (or simply *Up*) specifies the direction of counting.
 - ❖ $Up/Down = 1 \rightarrow$ Count upward
 - ❖ $Up/Down = 0 \rightarrow$ Count downward

Synchronous 3 bit Up/Down counter

- Up/Down Counter can either count up or down on each clock cycle
- Up counter counts from 000 to 111 and then changes back to 000
- Down counter counts from 111 to 000 and then back to 111
- Counter counts up or down each clock cycle
- Output changes occur on clock rising edge
- Number of Flip flop = 3

State Diagram



M is a Mode Selection line
M = 0, Up Counter
M = 1, Down Counter

M	Q ₃	Q ₂	Q ₁	Q ₃ *	Q ₂ *	Q ₁ *	T ₃	T ₂	T ₁
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	0	1	0	0	0	1

M Q ₃	Q ₂ Q ₁			
	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	1	0	0	0
10	1	0	0	0

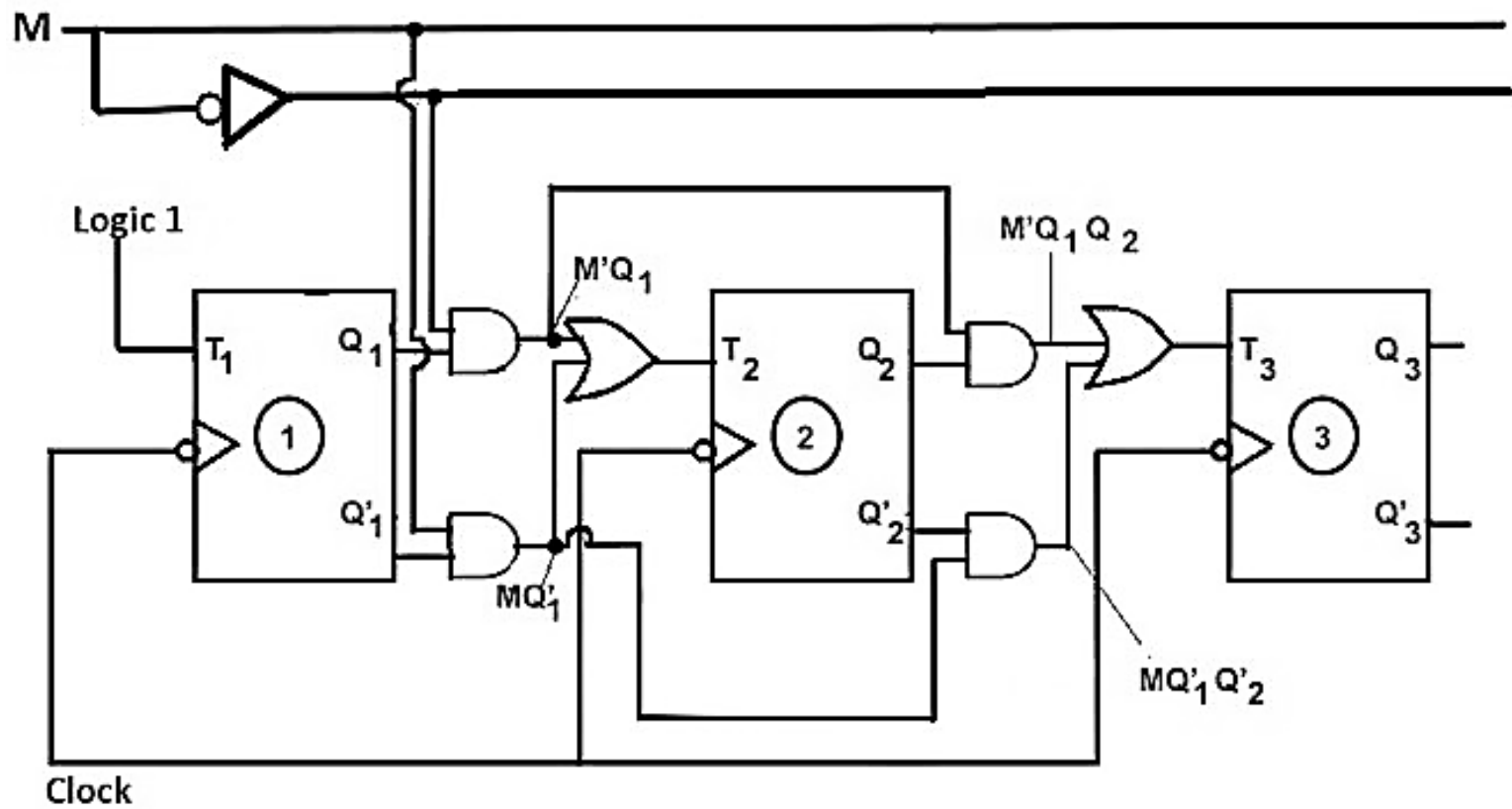
$$T_3 = M'Q_2Q_1 + MQ_2'Q_1'$$

M Q ₃	Q ₂ Q ₁			
	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	0	0	1
10	1	0	0	1

$$T_2 = M'Q_1' + MQ_1'$$

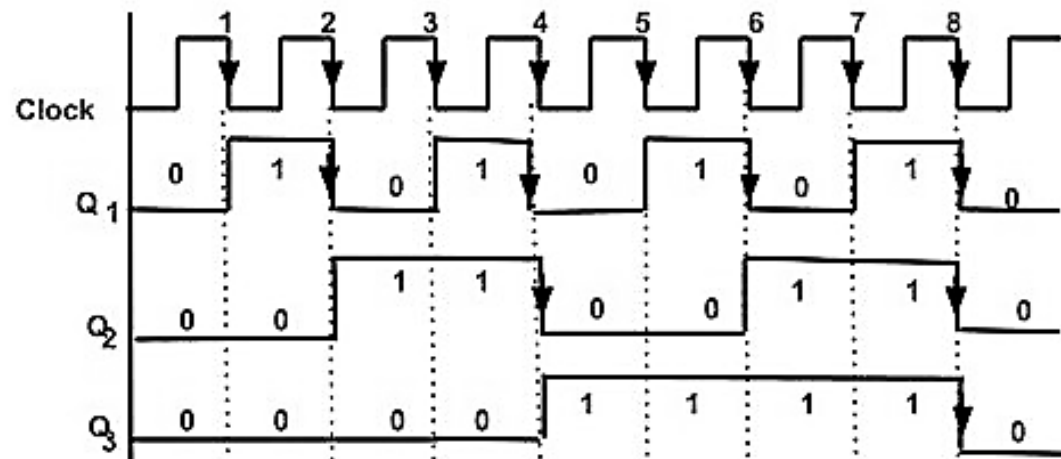
M Q ₃	Q ₂ Q ₁			
	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$T_1 = 1$$



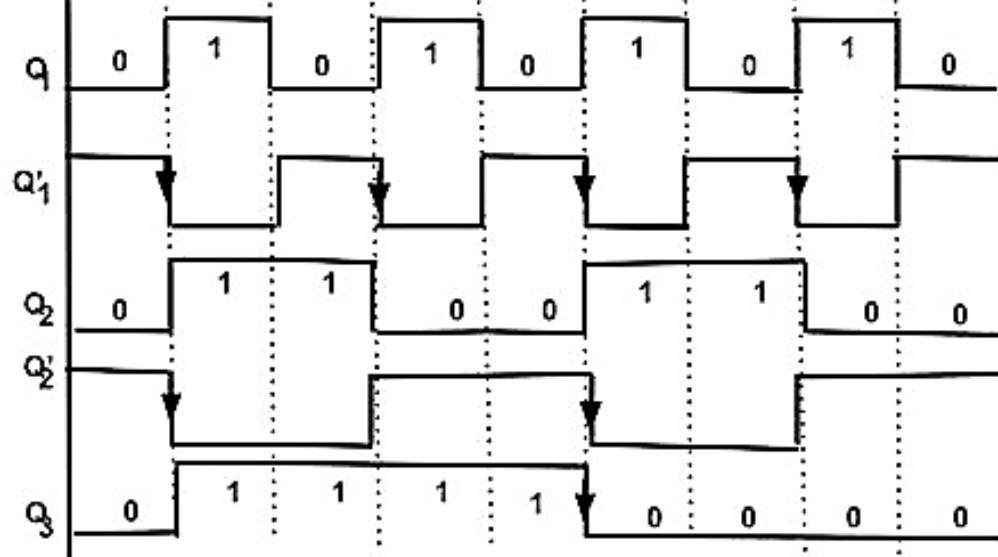
3 bit or Mod 8
Up counter

When $M = 0$



3 bit or Mod 8
Down counter

When $M = 1$



4 bit Up/Down Counter

```
module updowncounter (clk, rst, control,q);
```

```
    input clk, rst,  control;
```

```
    output [3:0]q;
```

```
    reg [3:0]q;
```

```
    always@(posedge clk)
```

```
    begin
```

```
        if(rst)
```

```
            q <= 4'b0000;
```

```
        else if(control)
```

```
            q <= q + 1;
```

```
        else
```

```
            q <= q - 1;
```

```
    end
```

```
endmodule
```

```
module updowncountertb();
```

```
    reg clk,rst,control;
```

```
    wire [3:0]q;
```

```
    updowncounter c1 (clk,rst,control,q);
```

```
    initial
```

```
    begin
```

```
        clk=0;  rst=1;  control=1;
```

```
    end
```

```
    always #50 clk = ~clk;
```

```
    initial
```

```
    begin
```

```
        #100 rst=0;
```

```
        #2000 control=0;
```

```
    end
```

```
endmodule
```

4 bit Up/Down Counter test bench

```
module updowncountertb();
  reg clk,rst,control;
  wire [3:0]q;
  updowncounter c1 (clk,rst,control,q);
  initial
  begin
    clk=0;  rst=1;  control=1;
  end
  always #50 clk = ~clk;
  initial
  begin
    #100 rst=0;
    #2000 control=0;
  end
end
endmodule
```

Verilog code of 8-Bit Up-Down Counter

```
module up_down_counter (out, up_down, clk, data, reset);
output [7:0] out;
input [7:0] data;
input up_down, clk, reset;
reg [7:0] out;
  always @(posedge clk)
    if (reset)          // active high reset
      begin
        out <= 8'b0 ;
      end
    else if (up_down)
      begin
        out <= out + 1;
      end
    else
      begin
        out <= out - 1;
      end
endmodule
```

Synchronous (Parallel) Mod - 10 / Decade / BCD counter

Clock pulse	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10 (recycle)	0	0	0	0

$$T_0 = 1$$

$$T_1 = Q_3' \cdot Q_0$$

$$T_2 = Q_1 \cdot Q_0$$

$$T_3 = Q_2 \cdot Q_1 \cdot Q_0 + Q_3 \cdot Q_0$$

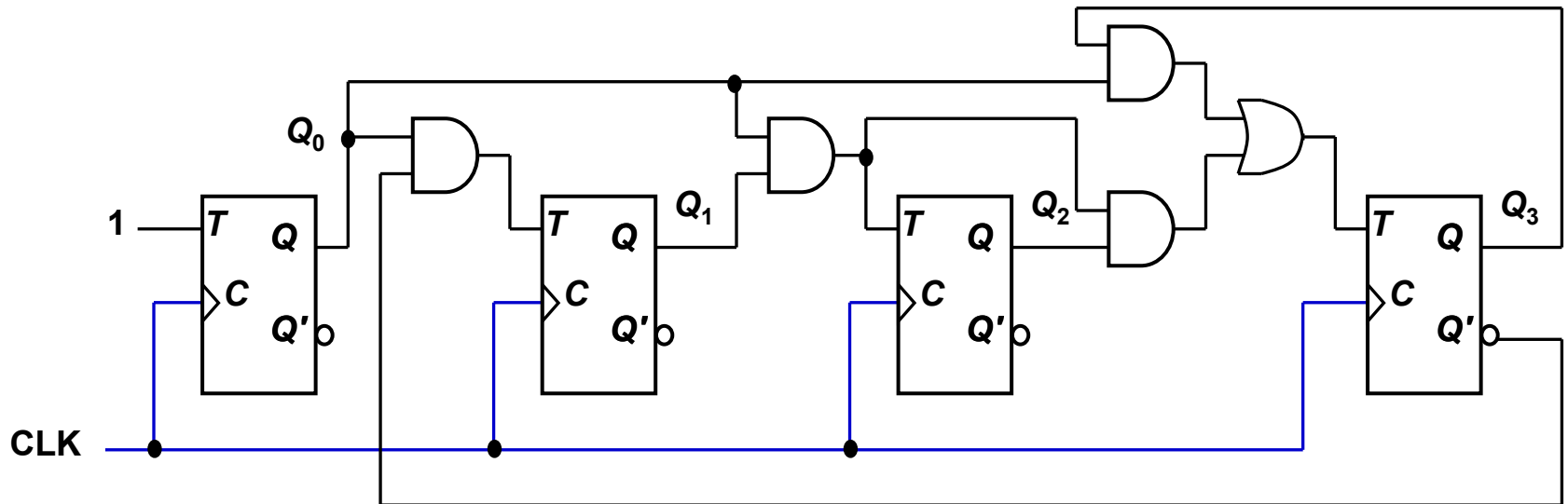
Synchronous (Parallel) Mod - 10 / Decade / BCD counter

$$T_0 = 1$$

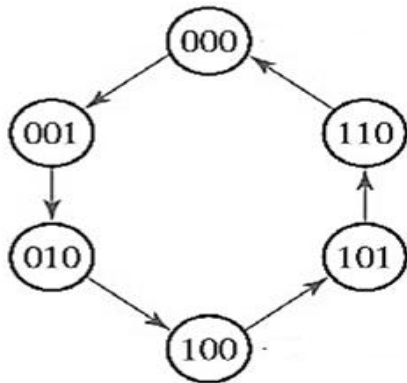
$$T_1 = Q_3' \cdot Q_0$$

$$T_2 = Q_1 \cdot Q_0$$

$$T_3 = Q_2 \cdot Q_1 \cdot Q_0 + Q_3 \cdot Q_0$$



Design counter (0,1,2,4,5,6)



Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

State Table for Counter

Present State			Next State			Flip-Flop Inputs					
A	B	C	A	B	C	J _A	K _A	J _B	K _B	J _C	K _C
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	1	0	0	1	X	X	1	0	X
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	0	0	0	X	1	X	1	0	X

Simplify using K-map
(unused states 3 & 7
as Don't care)

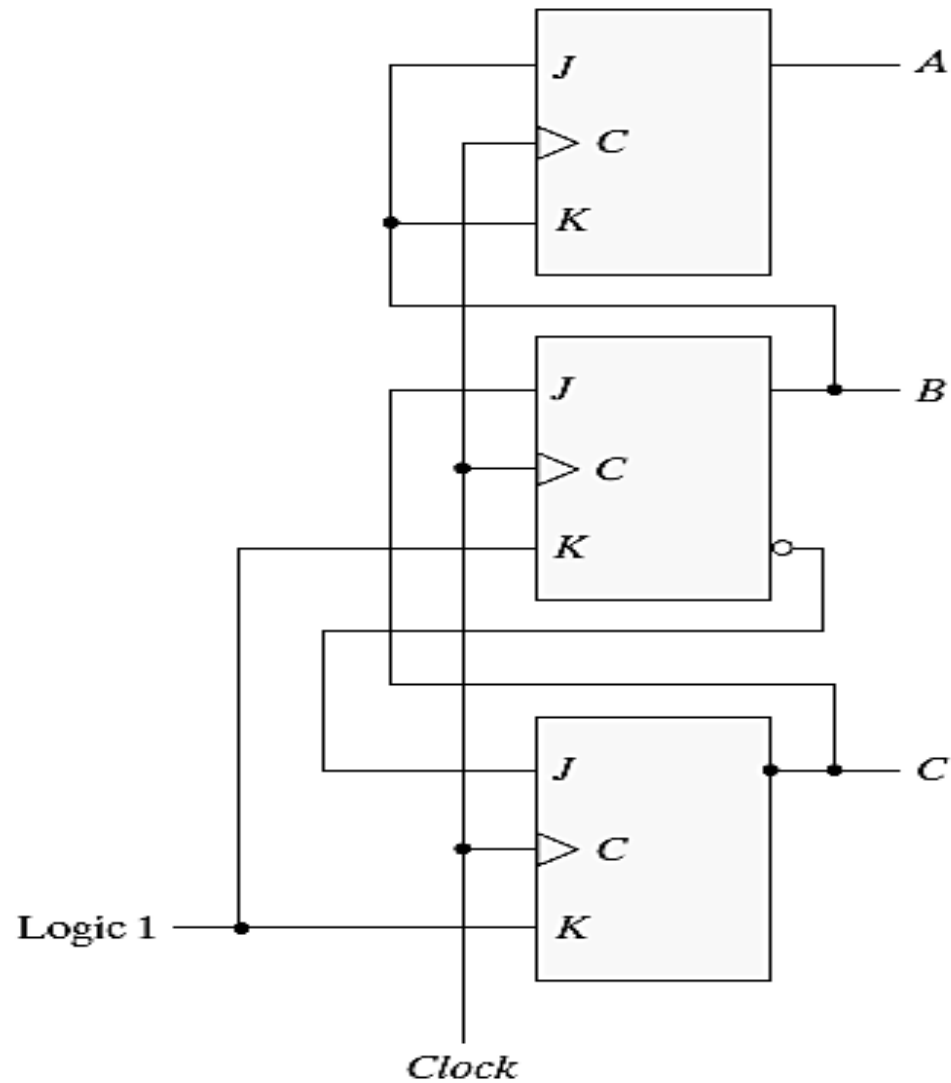
$$J_A = B \quad K_A = B$$

$$J_B = C \quad K_B = 1$$

$$J_C = B' \quad K_C = 1$$

Logic diagram

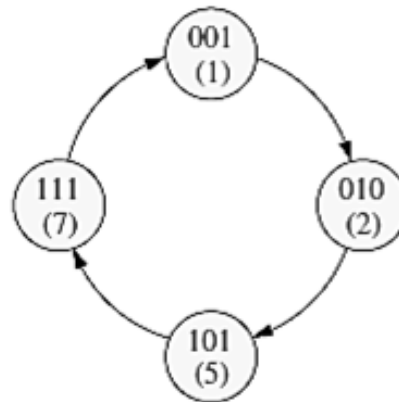
$$\begin{array}{ll} J_A = B & K_A = B \\ J_B = C & K_B = 1 \\ J_C = B' & K_C = 1 \end{array}$$



Design counter (1,2,5,7) using D FF

001 $\rightarrow$ 010 $\rightarrow$ 101 $\rightarrow$ 111

Step 1: State Diagram



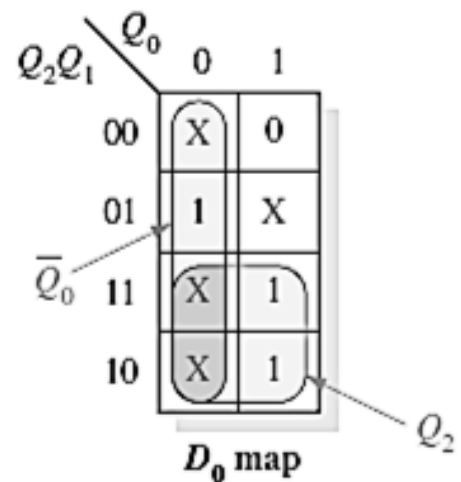
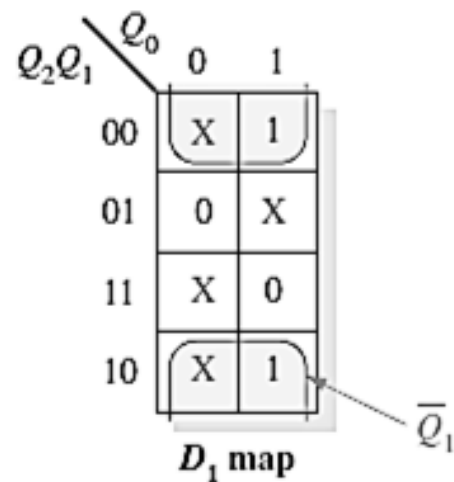
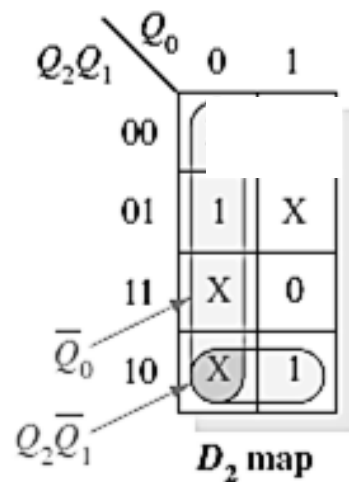
Step 2: Next-State Table

Present State			Next State		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	1	0	1	0
0	1	0	1	0	1
1	0	1	1	1	1
1	1	1	0	0	1

Step 3. Excitation table of D

Output Transitions			Flip-Flop Input
Q_N		Q_{N+1}	D
0	→	0	0
0	→	1	1
1	→	0	0
1	→	1	1

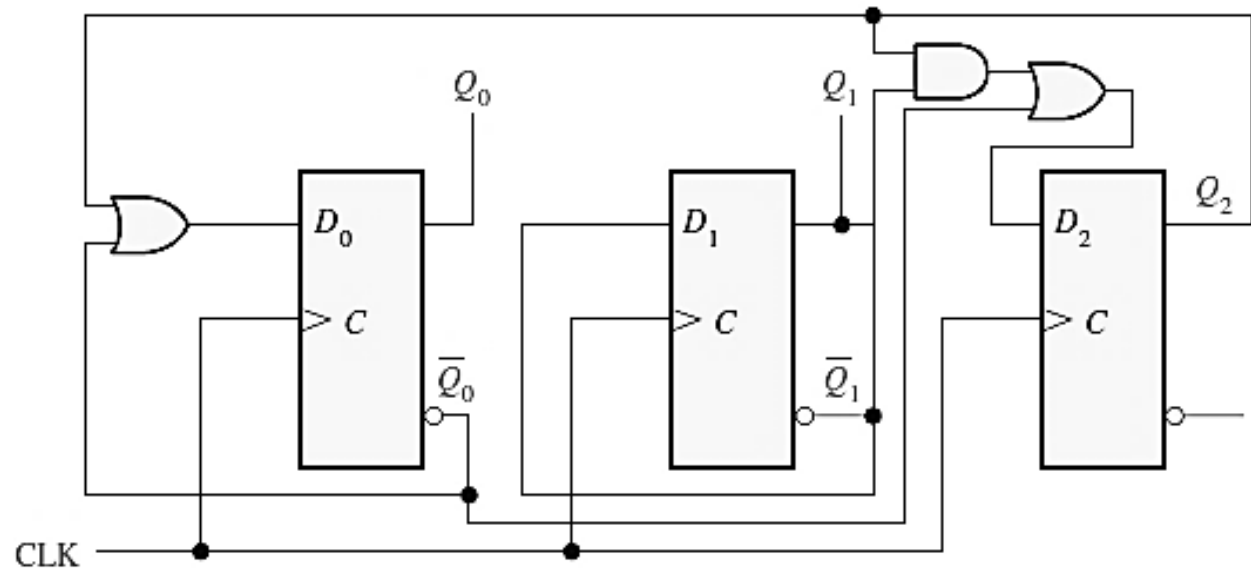
Present State			Next State					
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	D2	D1	D0
0	0	1	0	1	0	0	1	0
0	1	0	1	0	1	1	0	1
1	0	1	1	1	1	1	1	1
1	1	1	0	0	1	0	0	1



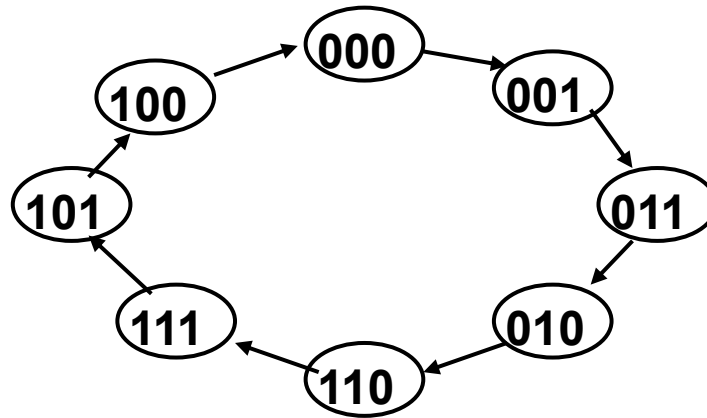
$$D_0 = \bar{Q}_0 + Q_2$$

$$D_1 = \bar{Q}_1$$

$$D_2 = \bar{Q}_0 + Q_2\bar{Q}_1$$



Design a 3-bit Gray code counter (using JK flip-flops)



Present state			Next state			Flip-flop inputs					
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	JQ_2	KQ_2	JQ_1	KQ_1	JQ_0	KQ_0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	1	0	X	1	X	X	0
0	1	1	0	1	0	0	X	X	0	X	1
0	1	0	1	1	0	1	X	X	0	0	X
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	1	0	1	X	0	X	1	X	0
1	0	1	1	0	0	X	0	0	X	X	1
1	0	0	0	0	0	X	1	0	X	0	X

		$Q_1 Q_0$			
		00	01	11	10
Q_2	0				1
	1	X	X	X	X

$$JQ_2 = Q_1 \cdot Q_0'$$

		$Q_1 Q_0$			
		00	01	11	10
Q_2	0	X	X	X	X
	1	1			

$$KQ_2 = Q_1' \cdot Q_0'$$

		$Q_1 Q_0$			
		00	01	11	10
Q_2	0		1	X	X
	1			X	X

$$JQ_1 = Q_2' \cdot Q_0$$

		$Q_1 Q_0$			
		00	01	11	10
Q_2	0	X	X		
	1	X	X	1	

$$KQ_1 = Q_2 \cdot Q_0$$

		$Q_1 Q_0$			
		00	01	11	10
Q_2	0	1	X	X	
	1		X	X	1

$$\begin{aligned} JQ_0 &= Q_2 \cdot Q_1 + Q_2' \cdot Q_1' \\ &= (Q_2 \oplus Q_1)' \end{aligned}$$

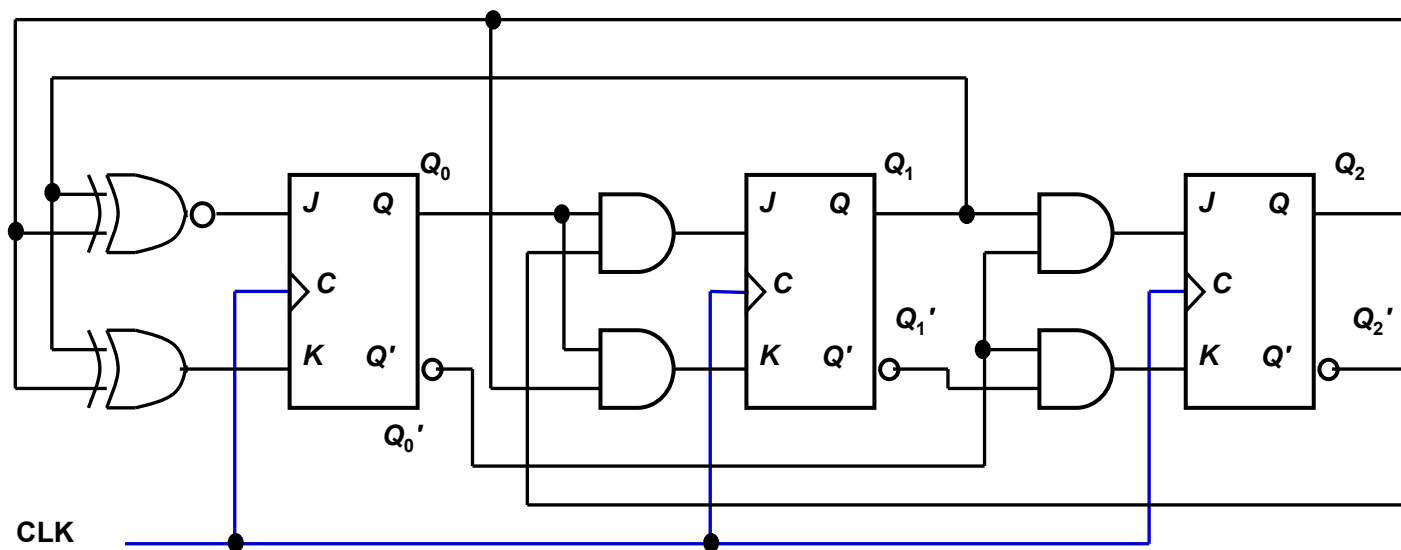
		$Q_1 Q_0$			
		00	01	11	10
Q_2	0	X		1	X
	1	X	1		X

$$\begin{aligned} KQ_0 &= Q_2 \cdot Q_1' + Q_2' \cdot Q_1 \\ &= Q_2 \oplus Q_1 \end{aligned}$$

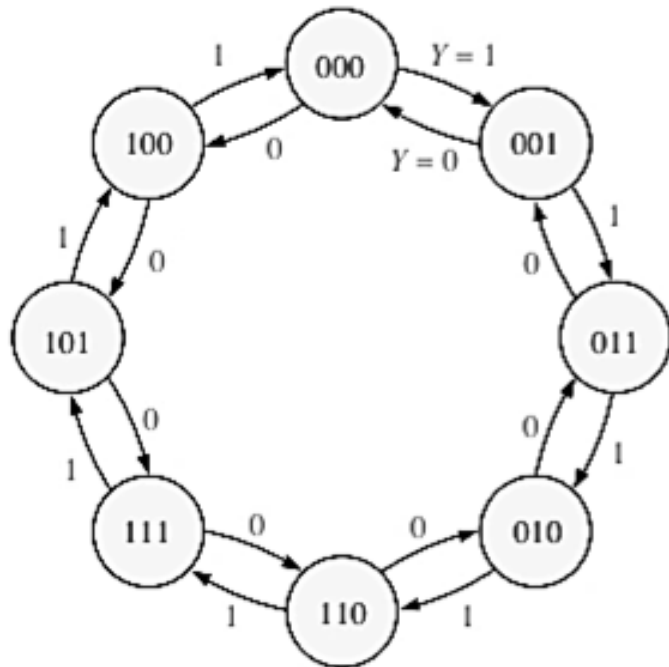
3-bit Gray code counter: logic diagram.

$$JQ_2 = Q_1 \cdot Q_0' \quad JQ_1 = Q_2' \cdot Q_0 \quad JQ_0 = (Q_2 \oplus Q_1)'$$

$$KQ_2 = Q_1' \cdot Q_0' \quad KQ_1 = Q_2 \cdot Q_0 \quad KQ_0 = Q_2 \oplus Q_1$$



Design a 3-bit Up/Down Gray code counter (using JK flip-flops)



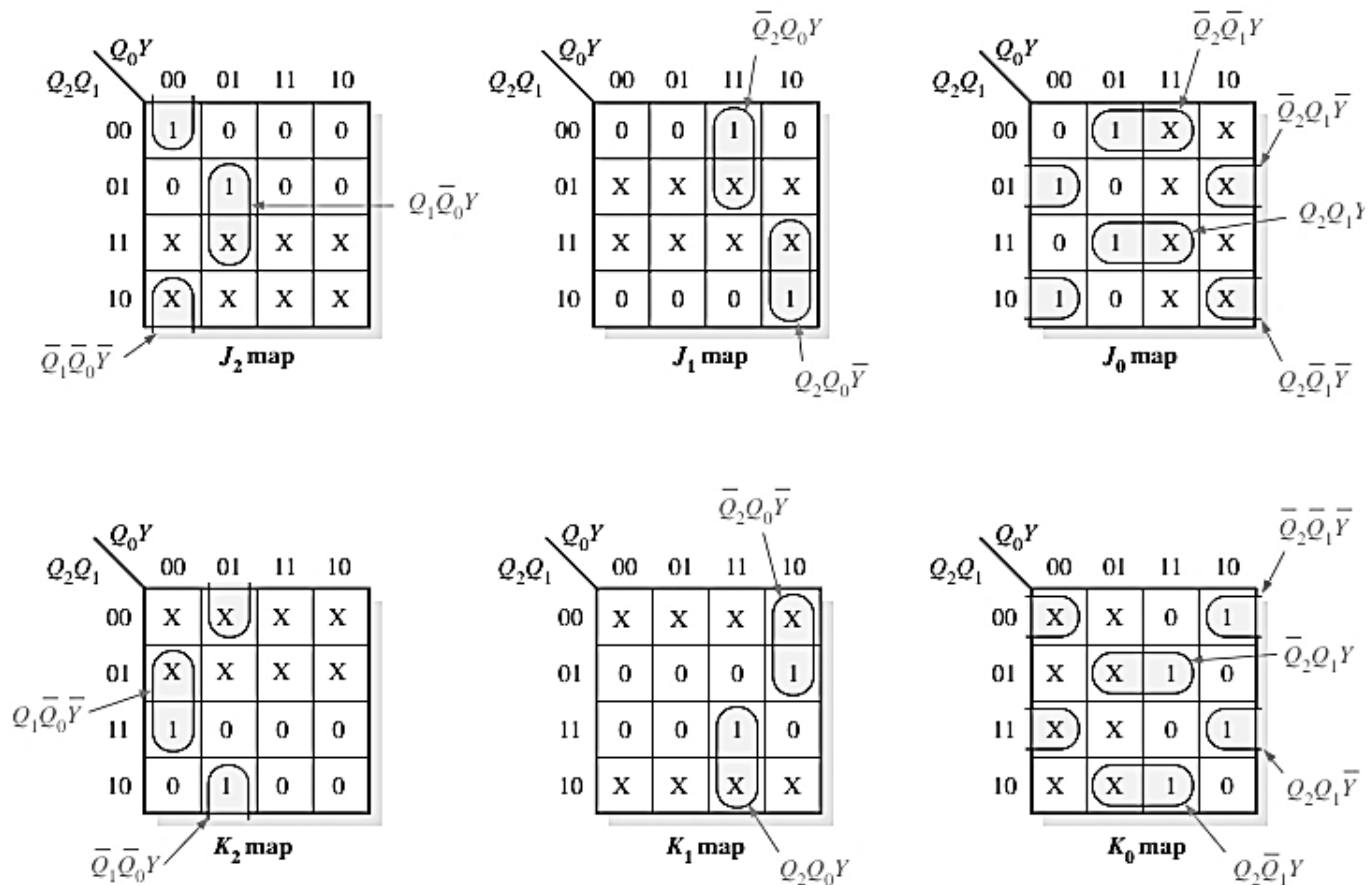
Next-state table for 3-bit up/down Gray code counter.

Present State			Next State					
			Y = 0 (DOWN)			Y = 1 (UP)		
			Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	0	1	0	0	0	0	1
0	0	1	0	0	0	0	1	1
0	1	1	0	0	1	0	1	0
0	1	0	0	1	1	1	1	0
1	1	0	0	1	0	1	1	1
1	1	1	1	1	0	1	0	1
1	0	1	1	1	1	1	0	0
1	0	0	1	0	1	0	0	0

$Y = \text{UP/DOWN}$ control input.

Transition table for a J-K flip-flop.

Output Transitions			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0



$$\begin{aligned}
 J_0 &= Q_2 Q_1 Y + Q_2 \bar{Q}_1 \bar{Y} + \bar{Q}_2 \bar{Q}_1 Y + \bar{Q}_2 Q_1 \bar{Y} & K_0 &= \bar{Q}_2 \bar{Q}_1 \bar{Y} + \bar{Q}_2 Q_1 Y + Q_2 \bar{Q}_1 Y + Q_2 Q_1 \bar{Y} \\
 J_1 &= \bar{Q}_2 Q_0 Y + Q_2 Q_0 \bar{Y} & K_1 &= \bar{Q}_2 Q_0 \bar{Y} + Q_2 Q_0 Y \\
 J_2 &= Q_1 \bar{Q}_0 Y + \bar{Q}_1 \bar{Q}_0 \bar{Y} & K_2 &= Q_1 \bar{Q}_0 \bar{Y} + \bar{Q}_1 \bar{Q}_0 Y
 \end{aligned}$$

Shift Register Counters

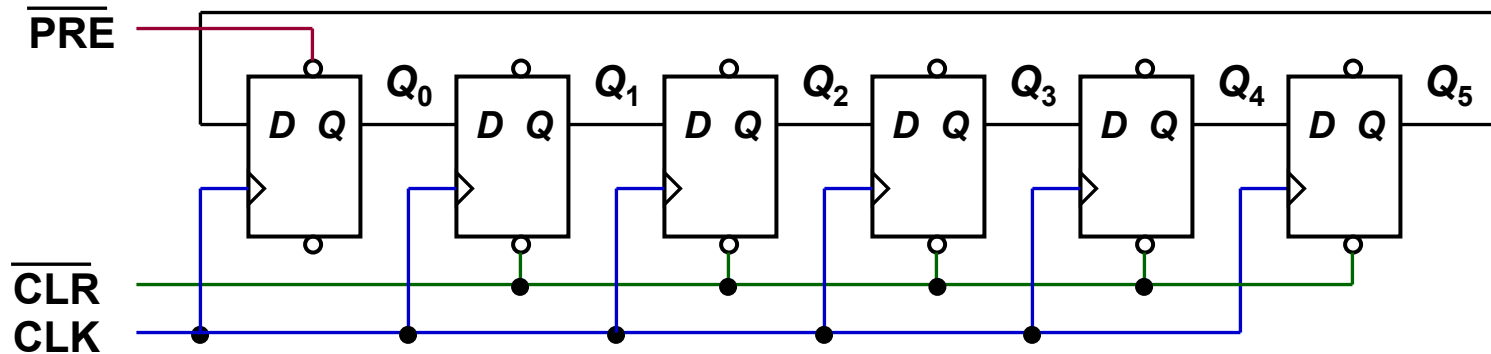
- **Shift register counter**: a shift register with the serial output connected back to the serial input.
- They are classified as counters because they give a specified sequence of states.
- Two common types: the *Johnson counter* and the *Ring counter*.

Ring Counters

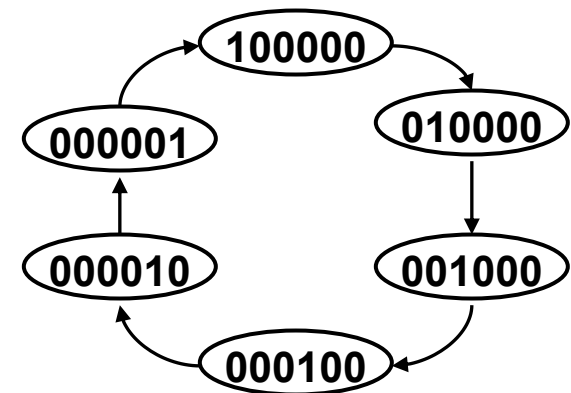
- One flip-flop (stage) for each state in the sequence.
- **The output of the last stage is connected to the D input of the first stage.**
- An n -bit ring counter cycles through n states.
- Number of states = Number of flip flops
- No decoding gates are required, as there is an output that corresponds to every state the counter is in.

Ring Counters

- Example: A 6-bit (MOD-6) ring counter.



Clock	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5
0	1	0	0	0	0	0
1	0	1	0	0	0	0
2	0	0	1	0	0	0
3	0	0	0	1	0	0
4	0	0	0	0	1	0
5	0	0	0	0	0	1

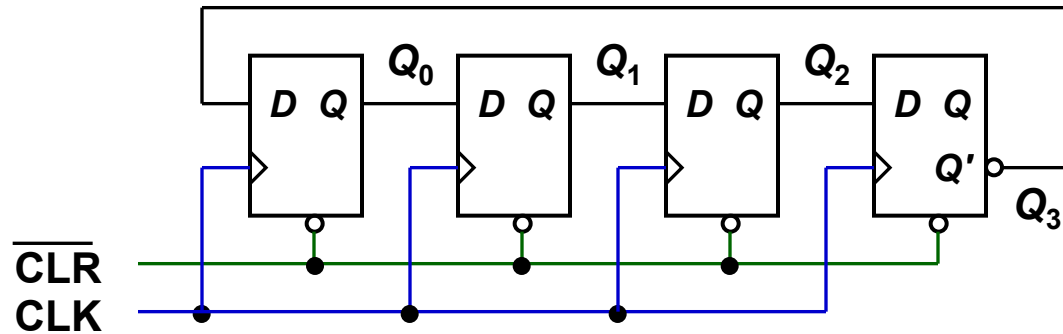


Johnson Counter / Twisted Ring Counter

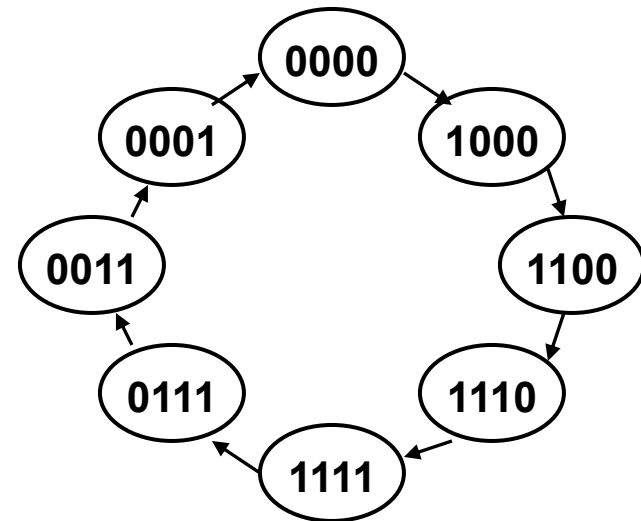
- The complement of the output of the last stage is connected back to the D input of the first stage.
- Also called the *twisted-ring / Switch tail ring counter*.
- Require fewer flip-flops than ring counters but more flip-flops than binary counters.
- An n -bit Johnson counter cycles through $2n$ states.
- Number of states = $2 \times$ number of flip flops.
- Require more decoding circuitry than ring counter but less than binary counters.

Johnson Counters

- Example: A 4-bit (MOD-8) Johnson counter.

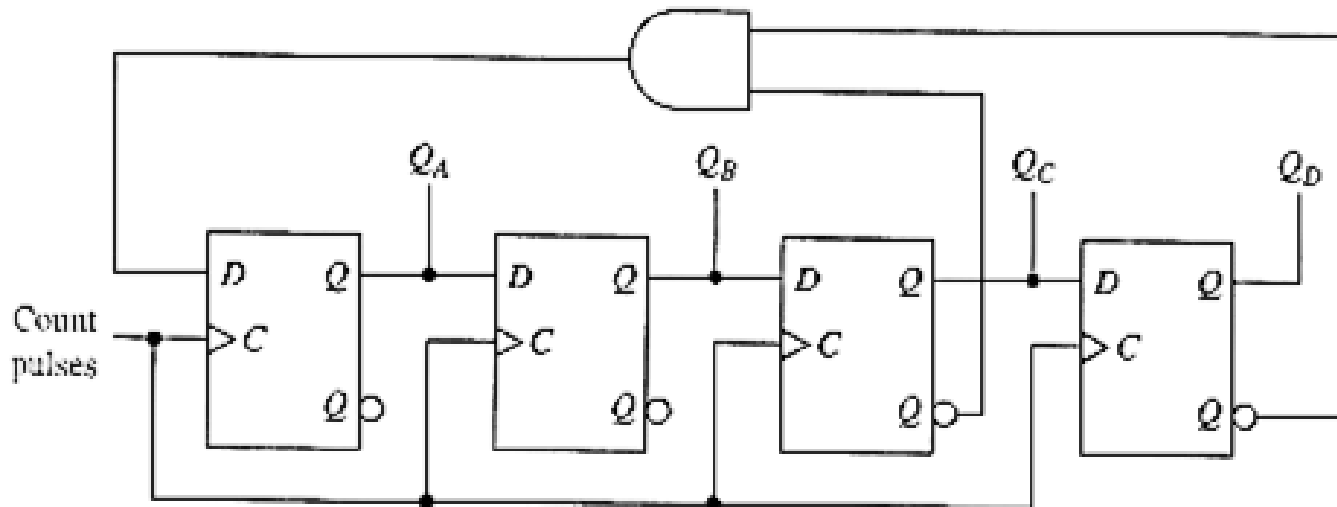


Clock	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1



MOD-7 Johnson Counters

- Twisted ring counter for odd number of states.
- The state consisting of all 1's is eliminated from the counting sequence.
- Achieved by connecting $\overline{Q}_C \overline{Q}_D$ to the input of the leftmost D flip-flop.
- Each state is detectable by use of a single two-input and-gate.



Q_A	Q_B	Q_C	Q_D
<u>0</u>	0	0	<u>0</u>
<u>1</u>	<u>0</u>	0	0
1	<u>1</u>	<u>0</u>	0
1	1	<u>1</u>	<u>0</u>
<u>0</u>	<u>1</u>	1	1
0	<u>0</u>	1	1
0	0	<u>0</u>	<u>1</u>
0	0	0	0
etc.			

